

DP Computer Science: Topic A4 Project Exemplar — Course Recommender

Overview

This exemplar project solution demonstrates a complete design for the "Course Recommender" mini-project. It shows how to apply computational thinking to design a system that recommends a full, valid IB DP subject package (3 HL, 3 SL) based on a user's profile.

Day 1: Problem Specification, Decomposition, Abstraction

1. Problem Specification

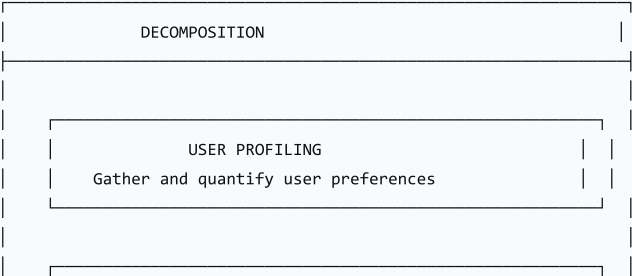
Element	Description
Problem Scenario	Prospective IB DP students must select a valid 6-subject package (3 HL, 3 SL) from the 6 subject groups. This is a complex decision that impacts university and career options. Students often lack a clear framework for mapping their personal interests and skills to an optimal subject combination.
Proposed Solution	A command-line tool that profiles a student based on their answers to a questionnaire about their academic interests. The system matches their profile to a predefined "career archetype" and recommends a valid 6-subject package.

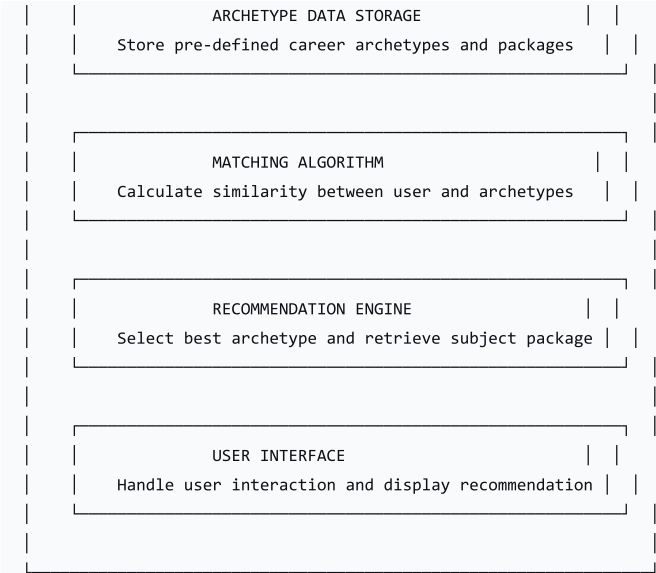
Success Criteria

#	Success Criteria
1	The system must ask the user to rate themselves on at least 5 different academic skill areas
2	The system must recommend a full, valid 6-subject package, specifying which 3 are HL and which 3 are SL
3	The recommendation must be generated by matching the user's profile to the closest of at least 4 pre-defined career archetypes using a distance-based algorithm

2. Decomposition

text





3. Abstraction

"A student's profile will be abstracted into a 6-dimensional vector based on their self-rated score (1-10) in key skill areas."

Feature	Description
Quantitative Reasoning	Solving complex maths problems
Scientific Inquiry	Conducting lab experiments, analysing data
Historical Analysis	Analysing past events and trends
Textual Interpretation	Analysing literature and texts
Creative Production	Creating art, music, or writing
Language Aptitude	Learning and using languages



4. Pattern Recognition

"The core pattern is that a user whose interest profile is 'closest' to a specific career archetype's profile is likely a good candidate for the subject package associated with that archetype. This is a nearest-neighbour classification problem."

Career Archetypes

Archetype	Vector	Subject Package
Future Engineer	[10, 9, 3, 4, 2, 5]	HL: Physics, Maths AA, Chemistry SL: English A, Spanish B, Economics
Future Diplomat	[5, 4, 9, 8, 6, 9]	HL: History, English A, Spanish B SL: Maths AI, Biology, Visual Arts
Future Doctor	[8, 10, 6, 5, 3, 7]	HL: Biology, Chemistry, Maths AA SL: English A, French B, Psychology
Future Artist	[2, 3, 5, 7, 10, 8]	HL: Visual Arts, English A, History SL: Maths AI, Biology, Language B

Day 2: Algorithm Design

Algorithm (K-NN with k=1)

```
text
|
|  ALGORITHM DESIGN
|
|  Data: Pre-defined dataset of career archetypes, each
|         with a name, an ideal vector, and a subject package.
|
|  Pseudocode:
|
|  FUNCTION recommend_package(user_profile_vector):
|      SET best_archetype to null
|      SET smallest_distance to infinity
|
|      FOR EACH archetype in dataset:
|          SET distance = calculate_distance(
|              user_profile_vector,
|              archetype.vector
|          )
|          IF distance < smallest_distance:
|              SET smallest_distance = distance
|              SET best_archetype = archetype
|
|      RETURN best_archetype.subject_package
|  END FUNCTION
|
```

Distance Calculation (Euclidean Distance)

```
python
def euclidean_distance(vec1, vec2):
    """Calculate the Euclidean distance between two vectors."""
```

```
return math.sqrt(sum([(a - b) ** 2 for a, b in zip(vec1, vec2)]))
```

Day 3: Code Implementation and Testing Plan

Python Code Implementation

python

```
import math
```

```
class SubjectPackageRecommender:
```

```
    """
```

```
    Recommends a full IB DP subject package based on user preferences.
```

```
    Matches the user to the closest career archetype using K-NN (k=1).
```

```
    """
```

```
def __init__(self):
```

```
    """Data: Career archetypes with ideal vectors and subject packages."""
```

```
    self.archetypes = {
```

```
        "Future Engineer": {
```

```
            "vector": [10, 9, 3, 4, 2, 5],
```

```
            "package": "HL: Physics, Maths AA, Chemistry\nSL: English A, Spanish B, Economics"
```

```
        },
```

```
        "Future Diplomat": {
```

```
            "vector": [5, 4, 9, 8, 6, 9],
```

```
            "package": "HL: History, English A, Spanish B\nSL: Maths AI, Biology, Visual Arts"
```

```
        },
```

```
        "Future Doctor": {
```

```
            "vector": [8, 10, 6, 5, 3, 7],
```

```
            "package": "HL: Biology, Chemistry, Maths AA\nSL: English A, French B, Psychology"
```

```
        },
```

```
        "Future Artist": {
```

```
            "vector": [2, 3, 5, 7, 10, 8],
```

```
            "package": "HL: Visual Arts, English A, History\nSL: Maths AI, Biology, Language B"
```

```
        }
```

```
    }
```

```
    self.questions = [
```

```
        "Rate your interest in Quantitative Reasoning (e.g., solving complex math problems): ",
```

```
        "Rate your interest in Scientific Inquiry (e.g., conducting lab experiments): ",
```

```
        "Rate your interest in Historical Analysis (e.g., analyzing past events): ",
```

```
        "Rate your interest in Textual Interpretation (e.g., analyzing literature): ",
```

```
        "Rate your interest in Creative Production (e.g., creating art, music, or writing): ",
```

```
        "Rate your interest in Language Aptitude (e.g., learning a new language): "
```

```
    ]
```

```
def get_user_profile(self):
```

```
    """Gets the 6-dimensional preference vector from the user."""
```

```
    user_profile = []
```

```
    print("\nPlease rate your interest on a scale of 1-10 for each area.")
```

```
    for question in self.questions:
```

```
        while True:
```

```
            try:
```

```
                answer = int(input(f"{question}"))
```

```
                if 1 <= answer <= 10:
```

```
                    user_profile.append(answer)
```

```
                    break
```

```
            else:
```

```
                print("Invalid input. Please enter a number between 1 and 10.")
```

```
        except ValueError:
```

```
            print("Invalid input. Please enter a number.")
```

```
        return user_profile

    @staticmethod
    def euclidean_distance(vec1, vec2):
        """Calculates the Euclidean distance between two vectors."""
        return math.sqrt(sum([(a - b) ** 2 for a, b in zip(vec1, vec2)]))

    def recommend(self, user_profile):
        """Finds the closest archetype and returns its subject package."""
        best_archetype_name = None
        min_distance = float('inf')

        for name, data in self.archetypes.items():
            distance = self.euclidean_distance(user_profile, data["vector"])
            if distance < min_distance:
                min_distance = distance
                best_archetype_name = name

        return self.archetypes[best_archetype_name]["package"]

def main():
    recommender = SubjectPackageRecommender()
    print("\n--- IB DP Subject Package Recommender ---")
    user_vector = recommender.get_user_profile()
    recommendation = recommender.recommend(user_vector)
    print("\n--- " * 10)
    print("Based on your profile, the recommended package is:")
    print(recommendation)
    print("\n")

if __name__ == "__main__":
    main()
```

Testing and Debugging Plan

Test Case	Input Data (Profile Vector)	Expected Output	Purpose of Test
1	[10, 10, 1, 2, 1, 3]	Future Engineer Package	Tests a clear-cut profile matches correctly
2	[7, 7, 7, 7, 7, 7]	Future Diplomat Package	Tests an ambiguous profile to verify distance calculation
3	User enters "eleven" for a rating	"Invalid input" error message	Tests input validation and robustness
4	[1, 1, 10, 9, 2, 8]	Future Diplomat Package	Tests historical/language profile
5	[2, 2, 4, 6, 10, 8]	Future Artist Package	Tests creative profile

Example Output

```
text

--- IB DP Subject Package Recommender ---

Please rate your interest on a scale of 1-10 for each area.
Rate your interest in Quantitative Reasoning (e.g., solving complex math problems): 10
```

Rate your interest in Scientific Inquiry (e.g., conducting lab experiments): 9
Rate your interest in Historical Analysis (e.g., analyzing past events): 3
Rate your interest in Textual Interpretation (e.g., analyzing literature): 4
Rate your interest in Creative Production (e.g., creating art, music, or writing): 2
Rate your interest in Language Aptitude (e.g., learning a new language): 5

Based on your profile, the recommended package is:
HL: Physics, Maths AA, Chemistry
SL: English A, Spanish B, Economics

Summary: Key Takeaways

Concept	Application in Project
Problem Specification	Defined clear success criteria
Decomposition	Broke the system into 5 components
Abstraction	Represented users as 6-dimensional vectors
Pattern Recognition	Identified nearest-neighbour classification
Algorithm Design	Used K-NN (k=1) with Euclidean distance
Testing	Developed multiple test cases

text

KEY TAKEAWAYS

✔

Archetypes simplify complex decisions

✔

K-NN provides a simple matching algorithm

✔

Euclidean distance works well for this problem

✔

Input validation is essential for robustness

✔

Multiple test cases ensure reliability

"This exemplar demonstrates how computational thinking concepts can be applied to design a practical recommendation system that helps students make informed decisions about their IB DP subject choices."